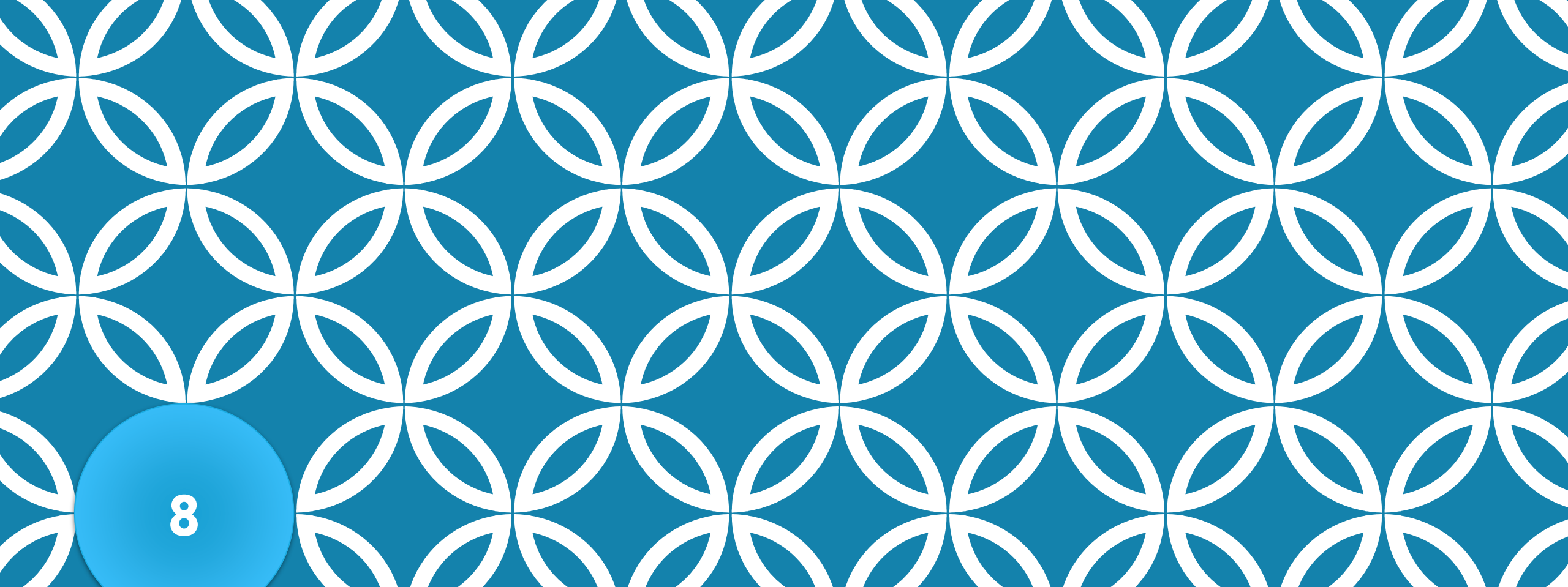




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Sequent calculus

Logic in Computer Science Lab
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Sequent Calculus or Gentzen Calculus

The sequent calculus is a method for constructing proofs to deduce whether or not a statement is true.

Sequent calculus always has a simple set of rules that is easy to check the validity of



Gerhard Karl Erich Gentzen

(1909–1945)

A German mathematician and logician.

DEFINITION

$$A_1, \dots, A_n \vdash B_1, \dots, B_k$$

The turnstile symbol divide the sequent into 2 clauses:

- On the left side of it we have **hypotheses**,
- On its right side we have **conclusions**;

Together them it denotes a **sequent**.

The turnstile can be thought of as a sort of implication.

$A \vdash A$ – You can read this sequent as “A implies A”

The $\vdash$ turnstile symbol

We can also use the $\rightarrow$ symbol.

Sources:

- Lectures of Prof. Tamás Kádek
- <http://logitext.mit.edu/tutorial>

A SEQUENT

$$\Gamma, A \vdash A, \Delta$$

Read: “Gamma **and** A imply A **or** Delta”

Important: the **comma means**

- **conjunction** on the **left** but
- **disjunction** on the **right side**.

Γ : hypotheses that are not relevant (convention)

Δ : conclusions which are not relevant (convention)

AXIOMS

The only **axioms** in this system are when some atomic clause appears on both sides of the turnstile, in general:

$$\Gamma, A \vdash A, \Delta$$

For example:

$$A \vdash A$$

$$A, B \supset C \vdash A, C \wedge A$$

VALIDITY

The sequel

$$A_1, \dots, A_n \vdash B_1, \dots, B_k$$

is valid, if at every ρ interpretation where

$$|A_1|_{\rho} = \dots = |A_n|_{\rho} = 1$$

there exist an i for that

$$|B_i|_{\rho} = 1$$

VALIDITY

The sequel

$$A_1, A_2, \dots, A_n \vdash B_1, B_2, \dots, B_m$$

is valid, if the form

$$\top \wedge A_1 \wedge A_2 \wedge \dots \wedge A_n \supset B_1 \vee B_2 \vee \dots \vee B_m \vee \perp$$

is valid.

EXERCISES

1. Give the formulas corresponding to the sequents!

(a) $\vdash (X \supset Y)$

(b) $X \vdash Y$

(c) $X \wedge Y \vdash Y \wedge Z$

(d) $(X \supset Y) \vdash$

(e) $X \vee Y \vdash Y \vee Z$

(f) $X, Y \vdash Y, Z$

Solutions:

a) $1 \supset (X \supset Y)$

b) $X \supset Y$

c) $X \wedge Y \supset Y \wedge Z$

d) $(X \supset Y) \supset 0$

e) $X \vee Y \supset Y \vee Z$

f) $X \wedge Y \supset Y \vee Z$

SEQUENT CALCULUS

The sequent calculus is a method for constructing proofs to deduce whether or not a statement is true.

Sequent calculus always has a simple set of rules that is easy to check the validity of

A **proof tree** is a proof if its every leaf is an axiom.

How can we construct proof trees ?

- By using **backward deduction**

Backward deduction is a logical inference, where we start with desired conclusion and work backwards to get rid of logical operators until we have atomic clauses.

DEDUCTION IN SEQUENT CALCULUS

A sequent is provable in the Gentzen calculus, if

- the sequent is an axiom,
- or exists a deduction rule, where the sequent in question is written under the line, and the sequents on the top of the line are provable.

In other words, if we want to prove a sequent in the Gentzen calculus which is not an axiom, it is enough to find a deduction rule, where the sequent in question is written below the line, and prove the sequents above the line.

(If everything above the line is true, so is everything under the line.)

PROOF TREES

Definition (Proof tree)

A **proof tree** is a tree whose nodes are sequents and where each parent-children fragment

$$\frac{S_1 \quad \dots \quad S_n}{S}$$

is an instance of a proof rule.

($\Rightarrow$ all leaves must be instances of axioms)

A sequent S is **provable** if there is a proof tree with root S .

Then we write $\vdash_G S$.

RULES

On the left side

On the right side

Equivalency:

$$(\equiv \rightarrow) \frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \equiv) \frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$$

Implication:

$$(\supset \rightarrow) \frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \supset) \frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$$

Conjunction:

$$(\wedge \rightarrow) \frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \wedge) \frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$$

Disjunction:

$$(\vee \rightarrow) \frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \vee) \frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$$

Negation:

$$(\neg \rightarrow) \frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$$

$$(\rightarrow \neg) \frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$$

PROOFS

The sequent calculus is a method for constructing proofs

A proof tree is a proof if its every leaf is an axiom

1. Sample

$$A \vdash A \wedge (A \vee B)$$

	On the left side	On the right side
$\equiv$	$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$
$\supset$	$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$
$\wedge$	$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$
$\vee$	$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$
$\neg$	$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$

Axiom: $A, \Gamma \rightarrow \Delta, A$

PROOFS

The sequent calculus is a method for constructing proofs

A proof tree is a proof if its every leaf is an axiom

1. Sample

$$\frac{A \vdash A \quad \frac{A \vdash A, B}{A \vdash A \vee B} (\vee r)}{A \vdash A \wedge (A \vee B)} (\wedge r)$$

The result: the sequent is axiomatically true

On the left side

On the right side

$$\equiv \frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$$

$$\supset \frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$$

$$\wedge \frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$$

$$\vee \frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$$

$$\neg \frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$$

Axiom: $A, \Gamma \rightarrow \Delta, A$

PROOFS

The sequent calculus is a method for constructing proofs

A proof tree is a proof if its every leaf is an axiom

2. Sample

$$X \rightarrow Y, \neg Y \vdash \neg X$$

	On the left side	On the right side
$\equiv$	$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$
$\supset$	$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$
$\wedge$	$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$
$\vee$	$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$
$\neg$	$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$

Axiom: $A, \Gamma \rightarrow \Delta, A$

PROOFS

The sequent calculus is a method for constructing proofs

A proof tree is a proof if its every leaf is an axiom

2. Sample

$$\begin{array}{c}
 \frac{X \vdash X, Y \quad X, Y \vdash Y}{X, X \rightarrow Y \vdash Y} (\rightarrow I) \\
 \frac{X, X \rightarrow Y \vdash Y}{X, X \rightarrow Y, \neg Y \vdash} (\neg I) \\
 \frac{X, X \rightarrow Y, \neg Y \vdash}{X \rightarrow Y, \neg Y \vdash \neg X} (\neg r)
 \end{array}$$

	On the left side	On the right side
$\equiv$	$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$
$\supset$	$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$
$\wedge$	$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$
$\vee$	$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$
$\neg$	$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$

Axiom: $A, \Gamma \rightarrow \Delta, A$

EXERCISE

Proof that this formula is a propositional law (i.e., it is true at every interpretation):

$$(X \vee Z) \wedge (Y \vee Z) \supset (X \wedge Y) \vee Z$$

$$(X \vee Z) \wedge (Y \vee Z) \supset (X \wedge Y) \vee Z$$

 $\equiv$
 $\supset$
 $\wedge$
 $\vee$
 $\neg$

On the left side

On the right side

$$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$$

$$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$$

$$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$$

$$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$$

$$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$$

$$\rightarrow (((X \vee Z) \wedge (Y \vee Z)) \supset ((X \wedge Y) \vee Z))$$

$$(X \vee Z) \wedge (Y \vee Z) \supset (X \wedge Y) \vee Z$$

 $\equiv$
 $\supset$
 $\wedge$
 $\vee$
 $\neg$

On the left side

On the right side

$$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$$

$$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$$

$$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$$

$$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

$$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$$

$$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$$

$$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$$

Axiom (Y)

Axiom (Z)

Axiom (Z)

Axiom (X)

Axiom (Z)

$$(X \vee Z), Y \rightarrow Z, Y$$

$$(X \vee Z), Z \rightarrow Z, Y$$

$$(X \vee Z), Z \rightarrow Z, X$$

$$Y, X \rightarrow Z, X$$

$$Y, Z \rightarrow Z, X$$

$$(X \vee Z), Y \rightarrow Z, X$$

$$(X \vee Z), (Y \vee Z) \rightarrow Z, Y$$

$$(X \vee Z), (Y \vee Z) \rightarrow Z, X$$

$$((X \vee Z) \wedge (Y \vee Z)) \rightarrow Z, Y$$

$$((X \vee Z) \wedge (Y \vee Z)) \rightarrow Z, X$$

$$((X \vee Z) \wedge (Y \vee Z)) \rightarrow (X \wedge Y), Z$$

$$((X \vee Z) \wedge (Y \vee Z)) \rightarrow ((X \wedge Y) \vee Z)$$

$$\rightarrow (((X \vee Z) \wedge (Y \vee Z)) \supset ((X \wedge Y) \vee Z))$$

The proof tree has 5 leaves;
every leaf is an axiom.
It means that we proved that
the sequent is valid, so
the original formula is valid.

YOU CAN USE SOME TOOLS

Try this: <http://logitext.mit.edu/main>

Logitext is an educational proof assistant online service

If we want to prove the form

$$X \supset Y, \neg Y \vdash \neg X$$

Then you have to write the following string in the field

$$X \rightarrow Y, \sim Y \mid - \sim X$$

$\mid -$ provable
 $\sim$ negation
 $\rightarrow$ implication
 $\wedge$ conjunction
 $\vee$ disjunction

The result:

$$\frac{\frac{\frac{\overline{X \vdash X, Y} \quad \overline{X, Y \vdash Y}}{X, X \rightarrow Y \vdash Y} (\rightarrow l)}{X, X \rightarrow Y, \neg Y \vdash} (\neg l)}{X \rightarrow Y, \neg Y \vdash \neg X} (\neg r)$$

EXERCISE

11.I.14. Proof that the following formula is a contradiction:

$$(a) \quad X \wedge \neg Y \wedge (\neg Y \supset \neg X)$$

	On the left side	On the right side
$\equiv$	$\frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$
$\supset$	$\frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$
$\wedge$	$\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$
$\vee$	$\frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$	$\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$
$\neg$	$\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$	$\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$